FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2015

ANSWERS, HINTS & SOLUTIONS

FULL TEST –V
(Main)

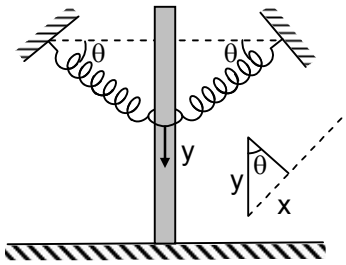
S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	D	A
2.	B	C	A
3.	C	A	B
4.	B	D	D
5.	A	C	B
6.	A	B	A
7.	A	D	C
8.	C	C	B
9.	A	C	D
10.	A	C	C
11.	A	A	C
12.	A	D	A
13.	B	B	B
14.	B	A	C
15.	A	B	D
16.	D	B	C
17.	B	A	C
18.	C	C	A
19.	C	D	C
20.	D	D	B
21.	A	C	B
22.	B	B	B
23.	B	D	C
24.	D	C	D
25.	D	B	B
26.	C	B	A
27.	B	C	A
28.	C	C	D
29.	B	B	B
30.	A	D	C

Physics

PART – I

SECTION – A

1. For small displacement $y = \frac{x}{\sin\theta}$



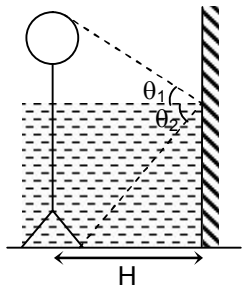
$2kx_0 \sin\theta = mg$ ..... (1)

If ring is displaced y

$mg - 2k\left(x_0 + y\sin\theta\right)\sin\theta = ma$

$a = -\frac{2k\sin^2\theta}{m}y$

2. $\mu_1 \cos\theta_1 = \mu_2 \cos\theta_2$



3. $W_F = \Delta U + \Delta K$ Length AC

$300\{1.5 - 0.9\} = \frac{1}{2}K\left(x_2^2 - x_1^2\right) + \frac{1}{2}mV^2 \quad \ell = \sqrt{(1.2)^2 + (0.9)^2}$

4. $\frac{1}{2}Ka^2 = \mu mg(a+b) + \frac{1}{2}Kb^2$

$(a-b) = 2\mu mg/K.$

5. $\tan\phi = \frac{\omega L}{R}$

$\cos\phi = \frac{R}{\sqrt{R^2 + X_L^2}}$

$i_0 = \frac{\varepsilon_0}{\sqrt{R^2 + X_L^2}}, \; V_L = IX_L = \frac{\varepsilon\, X_L}{\sqrt{R^2 + X_L^2}}$

6. $\nu = \frac{1}{2\pi\sqrt{LC}}$

7. $\frac{0 + V_A}{2} = V_1$

$\frac{V_A + V_B}{2} = V_2$

$\frac{V_B + 0}{2} = V_3$

8. $\frac{V_0 \times 6}{2} - \frac{\frac{V_0}{3} \times 2}{2} = 4$

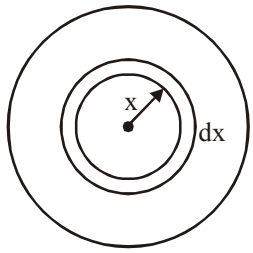
$V_0 = \frac{3}{2}$m/s

9. Kinetic energy at mean position $= 4J = \frac{1}{2}mA^2\omega^2 \Rightarrow \omega = 200$ Rad / s.

10. $mgh = \frac{1}{2}kx^2 - mgx$

11. $E = \frac{x}{2}\frac{dB}{dt}$

$E = \frac{3Kxt^2}{2}$



$d\tau = \frac{3Kxt^2}{2} \times \frac{2\pi x dx}{\pi r^2} q.x$

$\tau = \frac{3Kt^2q}{r^2}\int_0^r x^3 dx$

$\tau = \frac{3Kq.t^2}{4}.r^2$ …(i)

torque due to friction force

$d\tau = \mu dmgx$

$\tau = 2\mu g\frac{qm}{r^2}\int_0^r x^2 dx = \frac{2}{3}\mu mgr$ …(ii)

$\frac{3Kq.t^2r^2}{4} = \frac{2}{3}\mu mgr$

$t = \sqrt{\frac{8\mu mg}{9Kqr}}$ = 2 seconds.

12. About point O Ring is in pure rotation.

13. $\Delta\theta = W + \Delta U$

$\Delta U - \Delta Q - W$

14. Can be understood by graph.

15. $dQ = mS\Delta T < 0 \Rightarrow \Delta T < 0$

16. Light from any part of body can not reach the eye of man.

17. $a = \frac{g\sin\theta}{\left(1+\frac{I}{mr^2}\right)}$, $a_{ring} = \frac{g\sin\theta}{2}$

$a_{coin} = \frac{2}{3}g\sin\theta$

18. $F_{rat} = 0$

$F.2R - F.R = I\alpha \Rightarrow \alpha = 2.5\ Rad/s^2$

$\omega = \alpha t$

$K = \frac{1}{2} I\omega^2$

19. $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$7mgR = \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{2}{5}mv^2$

$mv^2 = 10mgR \Rightarrow v = \sqrt{10gR}$

20. Number of significant figures in the answer must be three.

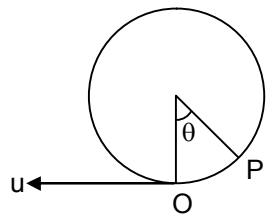
21. $\frac{T_1}{T_2} = a = 1$

$\frac{r_1}{r_2} = \frac{\sin\theta_1}{\sin\theta_2} = b = \frac{1}{\sqrt{3}}\ \frac{P_1}{P_2} = \frac{\cos\theta_1}{\cos\theta_2} = c = \sqrt{3}$

22. Escape velocity, $v = \sqrt{2gR}$

$\therefore \quad \frac{v_E}{v_M} = \sqrt{\frac{2g_E R_E}{2g_M R_M}} = \sqrt{6\times 4} = \sqrt{24}$

23. Tension in the string at the lowest point in vertical circle is given by



$T_L = mg + \frac{mu^2}{\ell}$

$2mg = mg + \frac{mu^2}{\ell}$

$u = \sqrt{g\ell}$

Using the law of conservation of energy, K.E. at O = P.E. at P

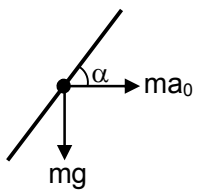
$\frac{1}{2}mu^2 = mg\ell(1-\cos\theta)$

$\frac{1}{2}mg\ell = mg\ell(1-\cos\theta)$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

24. Along the rod

$mg\sin\alpha - ma_0\cos\alpha = ma$



25. If the tension in the string is T then for insect
$T - mg = ma$

$T = \frac{3mg}{2}$

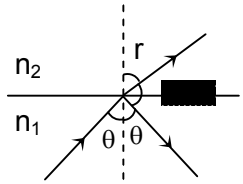
For block $T + N = 2mg$

26. For the velocity perpendicular to incline after collision will be same for both so time for both will remains.
For the velocity along the incline after collision, velocity of the ball on smooth surface will be greater than that of rough surface.

27. $mg(h+x) - F.x = 0$

28. $n_1 \sin\theta = n_2 \sin r = n_2 \sin(60 - \theta)$

$\sin c = \frac{n_2}{n_1} = \frac{\sin\theta}{\sin(60-\theta)} = \frac{2\sin\theta}{\sqrt{3}\cos\theta - \sin\theta}$



29. For bead to be stationary

slope of tangent on the curve $\frac{dy}{dx} = \tan\theta \le \mu$

$\frac{2a}{y} \le \mu.$

30. For each reaction involving both the processes, mass defect
$= 3 \times 2.014 - 4.001 - 1.007 = 0.026$ u

Energy released per deutron $= \frac{931.5 \times 0.026}{3}$ MeV

Let n be the number of deutrons required

Total energy $= \frac{931.5 \times 0.026}{3} \times n$ MeV $= \frac{P \times t}{\eta} = \frac{10^{16} \times 365 \times 24 \times 60 \times 60}{0.5}$

$n \simeq 5 \times 10^{35}.$

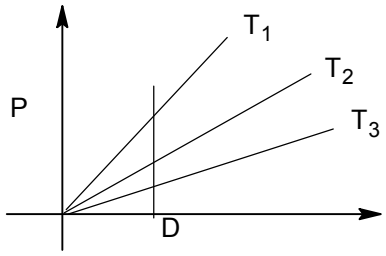
Chemistry

PART – II

SECTION – A

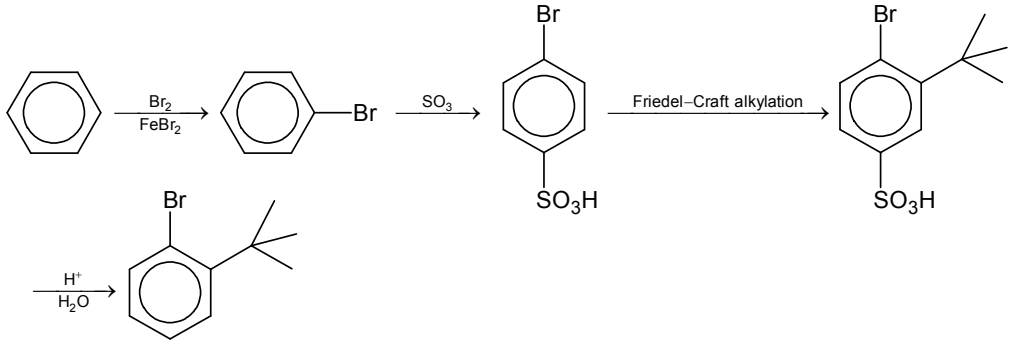
1. BaF_2 $\Delta H_{solution}$ = LE + $\Delta H_{hydratio}$
= 2367 – (457 ×2 + 1305)
= 148
BeF_2 $\Delta H_{solution}$ = 2906 – (457 × 2 + 2494)
= –502
For solubility : $\Delta H_{solution}$ should be negative.

3. At constant density, $P \propto T$

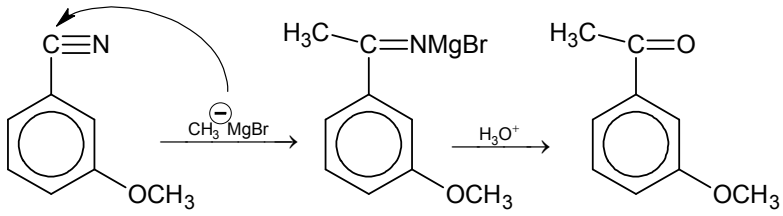


so, $T_1 > T_2 > T_3$

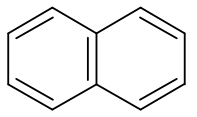
5.



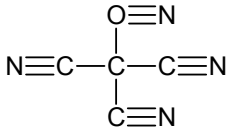
6.



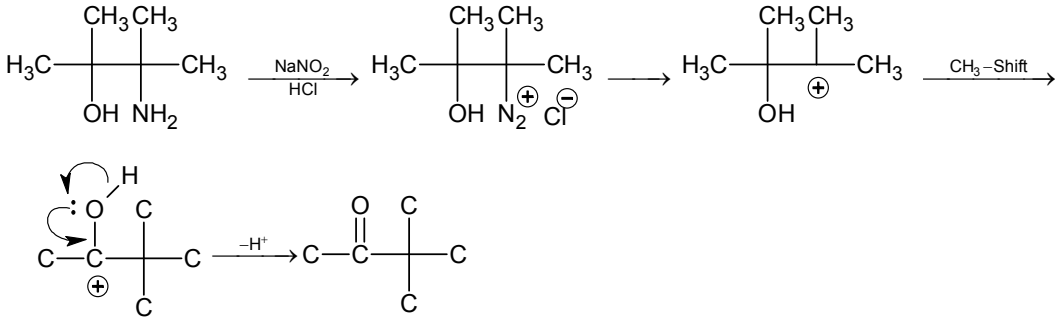
9.



19σ, 5π



8σ, 8π

14. 

15. $4HNO_3 \xrightarrow{h\nu} \underset{\text{Yellow}}{4NO_2} + O_2 + 2H_2O$

20. $V = \dfrac{W(\text{mass})}{d} = \dfrac{205.5}{0.79} = 260.13 \text{ ml}$

$M = \dfrac{W_B \times 1000}{m_B \times V} = \dfrac{5.5 \times 1000}{36.5 \times 260.13} = 0.58 \text{ M}$

22. According to Faraday's 1st law:

$W = \dfrac{QE}{F}$

$\text{Slope} = \dfrac{E}{F}$

27. $2XeF_6 + SiO_2 \longrightarrow 2XeOF_4 + SiF_4$

29. Hoffmann-degradation and Haloform reaction.

Mathematics

PART – III

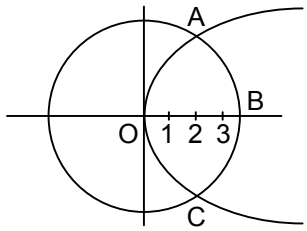
SECTION – A

1. $f(x) + 2f(1 - x) = 1 - x^2$ ..... (1)
$\Rightarrow 2f(x) + f(1 - x) = 1 - (1 - x)^2$ ..... (2)
$\Rightarrow f(x) = 1 - \frac{(x-2)^2}{3}$ {on solving (1) and (2)}
$\Rightarrow$ Range of f(x) is $(-\infty, 1]$

2. Required number of ways = $\frac{6!}{3! \cdot 3!} = 20$

3. Average = $\frac{2\sin 2^\circ + 6\sin 6^\circ + 10\sin 10^\circ + ... + 178\sin 178^\circ}{45}$
$= \frac{180\sin 2^\circ + 180\sin 6^\circ + ... + 90\sin 90^\circ}{45}$
$= \frac{180\left(\frac{\sin 44^\circ \sin 44^\circ}{\sin 2^\circ}\right) + 90}{45}$
$= 2\left(\frac{1 - \cos 88^\circ}{\sin 2^\circ} + 1\right)$
$= 2(\text{cosec } 2^\circ - 1 + 1)$
$\Rightarrow$ Average = 2 cosec 2°

4. Interior region is closed figure OABC
Integral co-ordinates will be corresponding to
x = 1, x = 2 and x = 3 only
Integral co-ordinates are (1, –1), (1, 0), (1, 1)
(2, –2), (2, –1), (2, 0), (2, 1), (2, 2), (3, –2), (3, –1)
(3, 0), (3, 1), (3, 2)
$\Rightarrow$ Total such points are 13



5. Projection = $\left|\frac{(3-1)\times 2 + (1-(-2))\times 1 + (0-3)\times 3}{\sqrt{2^2 + 1^2 + 3^2}}\right| = \left|\frac{4 + 3 - 9}{\sqrt{14}}\right|$
$\Rightarrow$ Projection = $\frac{2}{\sqrt{14}}$

6. $n = 2^6 = 64$
$m = 0 + 1 + 2 + 3 + 4 + 5 + 6 = 21$
$\Rightarrow n - m = 64 - 21 = 43$

7. $\cos 9x \cos 3x = \cos 18x \cos 12x$
$\Rightarrow \cos 12x + \cos 6x = \cos 30x + \cos 6x$
$\Rightarrow \cos 30x = \cos 12x$
$\Rightarrow 30x = 2n\pi \pm 12x$
$\Rightarrow 18x = 2n\pi, 42x = 2n\pi$
$\Rightarrow x = \frac{\pi n}{9}, x = \frac{\pi n}{21}$
Number of solutions = 10 + 22 – 4 = 28

($\because$ 0, $\frac{\pi}{3}$, $\frac{2\pi}{3}$, π will be common in both)

8. Let α and β be the roots of the equation $x^2 + px + q = 0$
$\Rightarrow \alpha + \beta = -p$, $\alpha\beta = q$
$\Rightarrow \alpha^3 + \beta^3 = -m$, $(\alpha\beta)^3 = n$
$\Rightarrow -m = \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$\Rightarrow -m = -p^3 + 3pq$
$\Rightarrow m = p^3 - 3pq$

9. Hyperbola will be of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
$2ae = 20$
$\Rightarrow ae = 10$, $\frac{b}{a} = \frac{4}{3}$
$\Rightarrow b = 4k$, $a = 3k$
$\Rightarrow b^2 = a^2(e^2 - 1)$
$\Rightarrow b^2 + a^2 = a^2e^2$
$\Rightarrow k = 2$
$\Rightarrow b = 8$, $a = 6$
$\Rightarrow \frac{x^2}{36} - \frac{y^2}{64} = 1$

10. Put $x = \tan\theta$

11. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 4 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 1(8+1) - 1(3) + 1(1-4)$ = 9 – 3 – 3 = 3

12. $|x - 1| + |x - 10| = 9 - \{x\}$
As LHS ≥ 9 and RHS ≤ 9
$\Rightarrow$ LHS = RHS = 9
$\Rightarrow \{x\} = 0 \ \forall \ x \in [1, 10]$
$\Rightarrow$ 10 solutions

13. Let x_1 and x_2 appears on the dice
$(x_1, x_2) \in \{(1, 3), (2, 4), (3, 5), (4, 6), (3, 1), (4, 2), (5, 3), (6, 4)\}$
$\Rightarrow$ Probability = $\frac{8}{6^2} = \frac{2}{9}$

14. The four sides of quadrilateral are given as x = 2, x = 6, y = 2, y = 6, which intersect at the points (2, 6), (2, 2), (6, 2), (6, 6)
The line which divides the area of this square in two equal parts will pass through its centre i.e. (4, 4)
$\Rightarrow y - 4 = \sqrt{3}(x - 4)$
$\Rightarrow y = \sqrt{3}(x - 4) + 4$

16. Let $x_1, x_2, \ldots\ldots, x_n$ be n observations, then

$\bar{X} = \frac{1}{2}\Sigma x_i$

Let $y_i = \frac{x_i}{\alpha} + 10$, then

$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{\alpha}\left(\frac{1}{n}\Sigma x_i\right) + \frac{1}{n}(10n) \Rightarrow \bar{Y} = \frac{1}{\alpha}\bar{X} + 10 = \frac{\bar{X} + 10\alpha}{\alpha}$.

17. $w^{13} = 1$

Now, $a + b = w + w^2 + w^3 + w^4 + w^5 + w^6 + w^{-6} + w^{-5} + w^{-4} + w^{-3} + w^{-2} + w^{-1}$

$= (w + w^2 + ... + w^6) + (w^7 + w^8 ..... w^{12}) = -1$

$ab = (w + w^3 + w^4 + w^{-4} + w^{-3} + w^{-1})(w^2 + w^5 + w^6 + w^{-6} + w^{-5} + w^{-2})$

$= 3(w + w^2 + w^3 + ... + w^{12}) = -3$

$\Rightarrow$ Quadratic equation with a and b as roots

$= x^2 - (a + b)x + ab = 0$

$\Rightarrow x^2 + x - 3 = 0$

18. Let M be $\left(\sqrt{18}\cos\theta, \sqrt{8}\sin\theta\right)$. As shortest distance lies along common normal

$\Rightarrow$ Given line will be parallel to the tangent at M

$\Rightarrow \frac{x\cos\theta}{\sqrt{18}} + \frac{y\sin\theta}{\sqrt{8}} = 1$ is parallel to $2x - 3y + 25 = 0$

$\Rightarrow -\frac{\sqrt{8}\cos\theta}{\sqrt{18}\sin\theta} = \frac{2}{3} \Rightarrow \tan\theta = -1$

$\Rightarrow \theta = \frac{3\pi}{4} \Rightarrow$ point M is (–3, 2)

20. As $b + c = 3a$

$\Rightarrow \frac{\sin B + \sin C}{\sin A} = 3$

$\Rightarrow \frac{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}} = 3$

$\Rightarrow \frac{\cos\frac{B}{2}\cos\frac{C}{2} + \sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}\cos\frac{C}{2} - \sin\frac{B}{2}\sin\frac{C}{2}} = 3$

$\Rightarrow \frac{\cot\frac{B}{2}\cot\frac{C}{2} + 1}{\cot\frac{B}{2}\cot\frac{C}{2} - 1} = 3$

$\Rightarrow \cot\frac{B}{2}\cot\frac{C}{2} = 2$

$\Rightarrow \tan\frac{B}{2}\tan\frac{C}{2} = \frac{1}{2}$

$\Rightarrow \left[\cot\frac{B}{2}\cot\frac{C}{2} + \tan\frac{B}{2}\tan\frac{C}{2}\right] = \left[2 + \frac{1}{2}\right] = 2$

21. $\cos^{-1}(\cos x) = x, \forall x \in (0, \pi]$
$\Rightarrow \sin^{-1}(\sin x) = 0 \Rightarrow x = \pi$ is the solution

22. f (x) is increasing function
So, $x^3 + x^2 + 2x - 1 > 2x + 1$
$x^3 + x^2 - 2 > 0$
$(x - 1)(x^2 + 2x + 2) > 0.$

23. Equation of normal to $(x - 3)^2 = 4(y + 1)$ at $(2t + 3, t^2 - 1)$ is $(x - 3) = -t(y + 1) + 2t + t^3$
As these normals pass through (h, k)
$\Rightarrow t^3 + (1 - k)t + (3 - h) = 0$
$\Rightarrow t_1 + t_2 + t_3 = 0, 2t_1 + 3 = 7, 2t_2 + 3 = -1$
$\Rightarrow t_1 = 2, t_2 = -2$
$\Rightarrow t_3 = 0$
$\therefore$ Point is (3, –1)

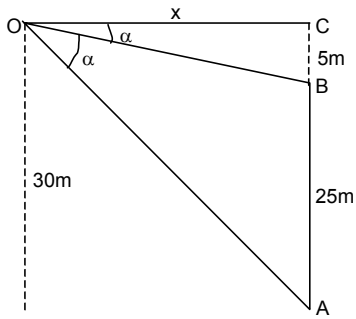
24. $2^{\alpha} + 2^{\beta} + 2^{\gamma} = 28$
$\Rightarrow \alpha, \beta \gamma$ can (2, 3, 4), (2, 4, 3), (3, 2, 4), (3, 4, 2), (4, 2, 3) and (4, 3, 2)

25. We have $\tan\alpha = \frac{5}{x}$ and $\tan 2\alpha = \frac{30}{x}$

$\Rightarrow \tan 2\alpha = \frac{30}{5\cot\alpha}$

$\Rightarrow \tan 2\alpha = 6\tan\alpha \Rightarrow 3 - 3\tan^2\alpha = 1$

$\Rightarrow \tan\alpha = \sqrt{\frac{2}{3}} \Rightarrow x = 5\cot\alpha = 5\sqrt{\frac{3}{2}}$ m

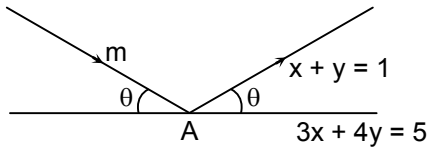


26. $$\sum_{r=0}^{9} \frac{1}{(2r+1)(2r+3)} = \sum_{r=0}^{9} \frac{1}{2}\left(\frac{(2r+3)-(2r+1)}{(2r+1)(2r+3)}\right)$$
$$\Rightarrow \sum_{r=0}^{9} \frac{1}{2}\left[\frac{1}{2r+1} - \frac{1}{2r+3}\right] = \frac{1}{2}\left[1 - \frac{1}{21}\right] = \frac{10}{21}$$

27. $$x^2\frac{dy}{dx} + 2xy + 2 + 6y\frac{dy}{dx} = 0$$
$$\Rightarrow \frac{d}{dx}\left(x^2y\right) + 2 + 6y\frac{dy}{dx} = 0$$
$\Rightarrow yx^2 + 2x + 3y^2 = c$

28. $$A + B = \frac{\pi}{4}$$
$\Rightarrow \cot(A + B) = 1$
$$\Rightarrow \frac{\cot A\cot B - 1}{\cot A + \cot B} = 1$$
$\Rightarrow (1 - \cot A)(1 - \cot B) = 2$

29. $\tan\theta = \dfrac{-\frac{3}{4} - m}{1 + \left(-\frac{3}{4}\right)m} = \dfrac{-1 - \left(-\frac{3}{4}\right)}{1 + (-1)\left(-\frac{3}{4}\right)}$



$\Rightarrow \dfrac{4m+3}{3m-4} = \dfrac{-\frac{1}{4}}{\frac{7}{4}}$

$\Rightarrow 7(4m + 3) = (4 - 3m)$

$\Rightarrow 31m = -17,\ m = -\dfrac{17}{31}$

A = (–1, 2)

$\Rightarrow$ Equation of line (L_1) is 17x + 31y = 45

30. m(m + 1)(m + 2) is product of three consecutive numbers, hence divisible by 6 and (7^m – 1) is divisible by 6. Hence, given number is always divisible by 6